Isaac Mubiru

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PROFESSIONAL SUMMARY

Backend engineer with 6+ years of production **JavaScript and Node.js**, specialising in **modelling complex distributed state** and **designing for failure**. Built a real-time two-way **sync between live MongoDB and Firestore stores** on Google Pub/Sub that carried parallel production traffic through a zero-downtime migration of 20+ applications, reporting to the CTO. Now applying that to **AI tooling**: customised and extended an open-source **agentic pipeline on Claude Code** with a deterministic verification gate in front of every LLM scoring step, and shipped a conversational assistant combining **OpenAI** with **Google Vision OCR** behind the WhatsApp Business API. Event-driven distributed systems background (Kafka, NATS, gRPC); works async across time zones.

CORE COMPETENCIES

Node.js & JavaScript | Distributed State & Sync | Designing for Failure (retries, cutovers) | Event-Driven Systems (Kafka, Pub/Sub, NATS) | LLM Integration (OpenAI, Vision OCR) | Agentic Workflows (Claude Code) | Microservices & gRPC | Remote, Async Delivery

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote

- Implemented a real-time **two-way sync** between MongoDB and Firestore using **Node.js and Google Pub/Sub** for message processing, keeping two live stores consistent while production traffic ran on both.
- Led the end-to-end migration of **20+ applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Saved the company **$5,000/month** by migrating all services off AWS; migrated file storage from Amazon S3 to GCS via a Python script.
- Built a 3D visualisation feature using video overlay and ffmpeg, increasing buyer conversion; implemented Express Shipping, adding **$40 revenue per order**.

MTailor

Mar 2022 – Jul 2022

Contractor · US Remote

- Built the migration script that read from MongoDB, ran complex processing logic and wrote to Firestore.
- Wrote documentation for fellow engineers on the new Firebase SDKs (Node.js and Python); trained the Ops team on the dashboard and backup procedures.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Extended backend APIs in **TypeScript** using Firebase Cloud Functions and Express.js hosted on Heroku.
- Built ad-hoc jobs querying over **2 million Firestore records** to keep customer prices current from Morningstar APIs; added responsive PWA pages in React and Tailwind.

Mind2matter.co

Jan 2022 – Jul 2022

Software Engineer · US Remote

- Built backends for DeFi applications using Web3 and Node.js; delivered React UIs from Figma designs under tight timelines, reporting to the CTO.

CodeBits

Jan 2020 – Jul 2021

Software Engineer · Uganda Remote

- Architected a **microservices backend** for FIDA Uganda's case management app using **Apache Kafka, Docker and Kubernetes**, with NATS Streaming and gRPC; led a team of 4 developers.
- Built a USSD service for legal aid providers communicating with a Node.js backend via **gRPC**.
- Developed a cross-platform React Native + Expo app (iOS & Android) with Firebase push notifications; built a paralegal database for LASPNET Uganda.

PROJECTS

Agentic Evaluation Pipeline AI tooling

Customised and extended an open-source agentic pipeline on Claude Code that scans job portals, confirms each posting is live in a headless browser before any LLM scoring, deduplicates sources and drafts tailored documents. Keeps a human on every irreversible step. 200+ evaluations, with verification-before-inference as the core design rule.

Claude Code · Playwright · Node.js · YAML

Healthcare WhatsApp Assistant LLM + vision

Conversational assistant that reads prescription photos and responds inside WhatsApp: OpenAI for dialogue, Google Vision for OCR, behind the WhatsApp Business API, handling dialogue state, media ingestion and platform limits.

OpenAI API · Google Cloud Vision · WhatsApp Business API · Node.js

Marketplace Recommendation Engine

Collaborative and content-based filtering engine in MongoDB for matching users to service providers.

Node.js · MongoDB

EDUCATION

B.Sc. Computer Engineering — Makerere University 2018 – 2022

Kampala, Uganda

SKILLS

Languages: JavaScript, TypeScript (~6 months, production), Python, Go (intermediate), Rust (intermediate)

Backend: Node.js, Express.js, REST APIs, gRPC, Microservices, Firebase Cloud Functions

AI / LLM: OpenAI API, Google Cloud Vision, Claude Code agentic workflows

Distributed & Data: Google Pub/Sub, Apache Kafka, NATS Streaming, MongoDB, Firestore

Cloud / Infra: GCP, Firebase, AWS (intermediate), Docker, Kubernetes, Linux, Git **Spoken:** English, Kiswahili